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NPS/003/001 – Technical Specification for 66kV and 132kV Voltage Transformers

1. Purpose

This document is the technical specification for 66kV and 132kV Voltage Transformers (VT s) for use on the CE Electric UK networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/003/001	1	Technical Specification for 66kV and 132kV Voltage Transformers

2. Scope

This specification details the requirements for 66 and 132kV Voltage Transformers (VT s) for use in open terminal substations on the CE Electric UK network. The VT s detailed in this specification are designed for protection, measurement and voltage control of power systems up to 132kV. This specification shall be used in conjunction with any project specific requirements detailed in Appendix 5, Addendum to Supplier Requirements.

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3 Technical Requirements

3.1 General Design Requirements

3.1.1 Voltage Transformers

VT s shall be in accordance with IEC 60044 Parts 2 and 5, the requirements listed below and in Appendix 1 of this document.

Two secondary outputs should be provided which shall be either protection or metering class (see Appendix 1).

VTs will be connected between line and earth in star configuration with the star point directly earthed.

VTs shall be either electromagnetic (low oil volume) or capacitor type.

The provision of arcing horns is required.

VTs will preferably have been assessed by the Energy Networks Association (ENA) and awarded an ENA Certificate of Conformance.

The technical specification of VTs must be confirmed using the table in Appendix 2.

3.1.2 Support Bushing

The support bushing of the VTs shall comply with the requirements of BS 3297-2 designation “C” .and Appendix 1 of this specification.

The support bushing shall be manufactured from grey silicon rubber hollow core composite material that have been tested in accordance with IEC 61462 and IEC 62217.

Hollow composite support bushings consist of an insulating tube manufactured from high strength glass fibre reinforced plastic bearing the mechanical load which is protected by an elastomeric housing.

3.1.3 Additional material requirements associated with the support bushing

Silicon rubber is a very generic term with many formulations. Variations in the composition of this material can adversely affect its long term performance to provide the following very important properties:-

UV resistance
Tracking Resistance
Hydrophobicity

As a result a range of good practice characteristic values have been included within this specification

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Material Composition	Minimum % per weight
Base Rubber consisting of approx 37% Silicone Polymer (Polydimethylsiloxane) (PDMS) and 10% Silica	47 %
Filler (ATH Aluminium tri hydrate and fumed silica)	52 %
Additional parts (pigments, crosslinkers)	1%

Where manufacturers differ from this characteristic composition, they shall provide supporting evidence about the long term experience of their product formulation.

The following table provides a list of characteristic properties that result from the above formulation. Manufacturers shall provide details of their characteristic properties for consideration.

Minimum acceptable values are detailed below.

Property	Minimum Value
Density	1.5g/cm
Passing Voltage Level of IEC 60587	4.5kV
Flammability Class of IEC 60695-11-10 of 3mm specimen	VO
Tensile Strength (Din 53504-S1)	6 N/mm2
Break Elongation (Din 53504-S1)	300%
UV resistance @ 300nm the energy of UV wavelength that equates to a molecular energy breakdown level of 398 kJ/mole *	445 kJ/Mole

* assumed wavelength of UV light (sun) 290 – 350nm

All silicon rubber insulators shall be manufactured using the HTV (high temperature vulcanising) method and shall ensure that the interface between the housing and the core is chemically bonded.

Flash or mould lines shall not exceed 1mm in height.

All glass fibre re-enforced plastic tubes shall be covered by an even concentric layer of silicon rubber that has a minimum insulation thickness of 3mm over the tube.

3.1.4 End Fittings

The end fittings transfer the mechanical load to the hollow tube core. The interface between the end fittings and the hollow tube shall be sealed by an elastomer with permanent elasticity. The sealing bond shall adhere to the surface of the metal fittings as well as the housing. Sealing by compression only is not acceptable.

3.1.5 Secondary Wiring Enclosure

The VT shall incorporate a waterproof Secondary Terminal Enclosure with a removable lid. This enclosure shall as a minimum comply with IP 65 in accordance with IEC 60529.

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All fixings used to secure the lid to the enclosure shall be captive.

The enclosure shall provide a means of terminating of an armoured protection cable using an M20 cable gland.

3.1.6 Design Life

VTs shall be designed to provide a minimum operating life of 40 years in normal outdoor service conditions as defined by IEC 62217:2006 clause 5, Table 1.

4. References

4.1 External Documentation

Reference	Title
BS EN 60044 - 2	Instrument Transformers – Part 2: Inductive Voltage Transformers
BS EN 60044 – 5	Instrument Transformers – Part 5: Capacitive Voltage Transformers
BS 3297-2	Characteristics of indoor and outdoor post insulators for systems with nominal voltages greater than 1000V
IEC 60529	Degrees of protection provided by enclosures (IP code)
BS 7354:1990	Code of practice. Design of high-voltage open terminal substations.
IEC 62231	Composite Station Post insulators for substations with a.c. voltages >1000V up to 245kV - – general definitions, test methods and acceptance criteria.
IEC 62217	Polymeric Insulators for indoor and outdoor use with a nominal voltage >1000V – general definitions, test methods and acceptance criteria.
IEC 61462	Composite hollow insulators – pressurised and un-pressurised insulators for use in electrical equipment with rated voltage greater than 1000V – Definitions, test methods, acceptance criteria and design recommendations
IEC 60587	Electrical Insulating materials used under severe ambient conditions – test methods for evaluating resistance to tracking and erosion
IEC 60695-11-10 Din 53504	Flammability classification Determination of tensile strength at break, tensile stress at yield, elongation at break and stress values of rubber in a tensile test

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4.2 Internal documentation

Reference	Title
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4.3 Amendments from Previous Version

Clause	Changes
3.1.2 – 3.1.4	The support bushings used on this product have been specified as silicon rubber hollow core composite insulators in place of the traditional ceramic designs. This provides manual handling improvements as the material is lighter than comparable porcelain. It also minimises the risk of material disruption in the event of a catastrophic failure.
Appendix 1	Modified to incorporate technical parameters associated with hollow tube composite insulators.
Appendix 4	Self certification conformance declaration schedules added.

5. Definitions

Term	Definition
Voltage Transformer	An instrument in which the secondary voltage, in normal conditions of use, is substantially proportional to the primary voltage, in accordance with the specified ratio and which differs in phase from the primary voltage by an angle which is approximately zero for an appropriate direction of the connections.

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G. Hammel	Overhead Standards Manager	G. Hammel	21/04/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	21/04/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Standards Plant Manager	D Blackledge	22/04/2010

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	CDS Administrator	S.Johnson	27/04/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	21/04/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	22/04/2010

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Appendix 1: Electrical Ratings 66kV and 132kV VTs

66kV			
	BS EN 60044 Part 2 Reference	BS EN 60044 Part 5 Reference	
Nominal System Voltage			66 kV
Highest Voltage for Equipment U_m (rms)	Table 4	Table 4 Range I	72.5kV
Rated Frequency			50Hz
Design			Outdoor Post Type VT,
Ambient air temperature range	4.1.1	5.1.1	-25/40 °C
System earthing	4.3	5.3	Solidly Earthed Neutral System.
Rated Secondary Voltage	5.1.2	6.2.2	110/√3 V
Ratios			66000/√3 : 110/√3
Number of secondary outputs			2 (Protection, metering, or dual purpose as specified at the time of ordering)
Class of Secondary Output Types			Metering: Class 0.5 Protection: Class 3P Dual purpose: Class 3P / 0.5
Output	5.2	6.3	50VA per winding
Voltage Factor	5.3	6.4	1.2 Continuous 1.5 for 30 sec
Limits of Temperature Rise <i>To be limited by the lowest class of insulation either of the winding or the surrounding medium – manufactures to specify</i>	5.4	6.5	In accordance with Table 3 or manufactures data for specific material types
Rated short time thermal current			Minimum 25kA for 3 secs but not less than that of associated HV Switchgear
Partial Discharge – permissible levels (PD levels in Pc)	6.1.2.3 Table 7	7.2.3 Table	Values associated with earthed neutral systems
Insulation requirements for secondary windings	6.1.4	7.2.7.3	3kV (r.m.s) for 60 secs
Impulse Withstand Level:	6.1.1	7.1	325kV
Power Frequency Withstand Voltage	6.1.1	7.1	140kV
External Insulation Requirements: Pollution level	6.1.5.1	7.2.8	Level 3: Heavy

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Appendix 1: Electrical Ratings 66kV and 132kV VTs (cont'd)

Insulator Bushing Requirements		
Insulation Type		Composite Insulation (Silicon Rubber)
Rated insulation levels of the primary winding – (Wet 1min PF Withstand)	5.1.1.2 Table 3	140 kV
Rated insulation levels of the primary winding - (peak) – Dry Impulse	5.1.1.2 Table 3	325 kV
Length of Post Insulators (min)		770mm +/- 1mm
Minimum phase to earth creepage distance Dm<300mm		1600mm
External Insulation Requirements: Pollution level – creepage distance	5.1.6.1	Level 3: 25mm/kV (Heavy)
Mechanical Requirement for CT	5.2	Load class II: Static withstand load 2.5kN
Mechanical requirement for support insulator	BS3297 clause 4 (type D)	Strength class 4 (4kN)
Insulator Base Fixing Arrangement		Where the base of the CT support insulator mounted directly onto support steelwork the PCD shall be as stated. (Where an additional under base is used, the fixings shall be agreed with the purchaser)
Pitch circle diameter (mm)	127	
Bolt Hole (tapped hole)	4 x M16	
Metal Fittings (internal/external)	External	

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Appendix 1: Electrical Ratings 66kV and 132kV VTs (cont'd)

<u>132kV</u>			
	BS EN 60044 Part 2 Reference	BS EN 60044 Part 5 Reference	
Nominal System Voltage			132 kV
Highest Voltage for Equipment U_m (rms)	Table 4	Table 4 Range I	145kV
Rated Frequency			50Hz
Design			Outdoor Post Type VT,
Ambient air temperature range	4.1.1	5.1.1	-25/40 °C
System earthing	4.3	5.3	Solidly Earthed Neutral System.
Rated Secondary Voltage	5.1.2	6.2.2	110/√3 V
Ratios			132000/√3 : 110/√3
Number of secondary outputs			2 (Protection, metering, or dual purpose as specified at the time of ordering)
Class of Secondary Output Types			Metering: Class 0.5 Protection: Class 3P Dual purpose: Class 3P / 0.5
Output	5.2	6.3	50VA per winding
Voltage Factor	5.3	6.4	1.2 Continuous 1.5 for 30 sec
Limits of Temperature Rise <i>To be limited by the lowest class of insulation either of the winding or the surrounding medium – manufactures to specify</i>	5.4	6.5	In accordance with Table 3 or manufactures data for specific material types
Rated short time thermal current			Minimum 31.5kA for 3 secs but not less than that of associated HV Switchgear
Partial Discharge – permissible levels (PD levels in Pc)	6.1.2.3 Table 7	7.2.3 Table	Values associated with earthed neutral systems
Insulation requirements for secondary windings	6.1.4	7.2.7.3	3kV (r.m.s) for 60 secs
Impulse Withstand Level:	6.1.1	7.1	650kV
Power Frequency Withstand Voltage	6.1.1	7.1	275kV
External Insulation Requirements: Pollution level	6.1.5.1	7.2.8	Level 3: Heavy

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Appendix 1: Electrical Ratings 66kV and 132kV VTs (cont'd)

Insulator Bushing Requirements		
Insulation Type		Composite Insulation (Silicon Rubber)
Rated insulation levels of the primary winding – (Wet 1min PF Withstand)	5.1.1.2 Table 3	275 kV
Rated insulation levels of the primary winding - (peak) – Dry Impulse	5.1.1.2 Table 3	650 kV
Length of Post Insulators (min)		1500 +/- 2.5mm
Minimum phase to earth creepage distance		3350mm
External Insulation Requirements: Pollution level – creepage distance	5.1.6.1	Level 3: 25mm/kV (Heavy)
Mechanical Requirement for CT	5.2	Load class II: Static withstand load 3kN
Mechanical requirement for support insulator	BS3297 clause 4 (type D)	Strength class 4 (4kN)
Insulator Base Fixing Arrangement		Where the base of the CT support insulator mounted directly onto support steelwork the PCD shall be as stated. (Where an additional under base is used, the fixings shall be agreed with the purchaser)
Pitch circle diameter (mm)	127	
Bolt Hole (Tapped Hole)	4 x M16	
Metal Fittings (internal/external)	External	

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Appendix 2: Declaration of technical performance of VT

MANUFACTURERS TYPE REFERENCE			
Rating	Units	66kV System Voltage	132kV System Voltage
Equipment Design Voltage			
Rated Frequency	Hz		
Ambient Air Temperature Range	°C		
Service Conditions – Suitable for outdoor use?			
Suitable for solidly earthed system?			
Limits of Temperature Rise – and or Insulation class	°C		
WVT / CVT			
Ratio			
Number of Secondary Outputs			
Class of Secondary Outputs			
Highest Voltage of Equipment	kV		
Impulse Withstand Level	kV		
Power Frequency Withstand Voltage	kV		
Rated Frequency	Hz		
Rated Voltage Factor: Continuous Short Time			
Rated capacitance of the capacitor divider	pF		
Rated capacitance of the high voltage capacitor	pF		
Rated capacitance of the intermediate voltage capacitor	pF		
Number of capacitor units			
Oil (Capacitor divider) Type Mass	kg		
Oil (Magnetic unit) Type Mass	kg		
Weight of Complete VT	kg		
Standard edition and year			
Current I: Connection A1-A2	A		
Voltage <i>U_{pr}</i>	kV		
<i>U_{sr}</i> (V): Rated Secondary Voltage for each secondary winding. Protection Metering Dual Purpose	V		
Values of rated output Protection	VA		

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MANUFACTURERS TYPE REFERENCE			
Rating	Units	66kV System Voltage	132kV System Voltage
Metering Dual Purpose			
Accuracy Class: Protection Metering Dual Purpose			
Maximum simultaneous output for windings of a complete CVT regarding the accuracy class Protection Metering Protection + Metering			
External Insulation Requirements: Pollution level	Class / KV/mm		
HV Insulator creepage distance	mm		
HV Insulator minimum arcing distance	mm		
Length of Insulator	mm		
Typical PD Values	Pc		
Winding insulation medium			
Static mechanical withstand load	N		
Insulation material on CT			
HV Post Insulator material			
Bolt sizes for connection (HV & LV)			
Weight of complete Unit			
Details of mounting arrangements Flange mounting plate PCD details			
Has the equipment been awarded an ENA certificate of performance? Please provide reference number.			

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

66 & 132kV Inductive or Capacitive Voltage Transformers shall comply with the latest issues of the relevant national and international standards.

This check sheet identifies the clauses of IEC 60044-2, IEC 60044-5, IEC 61462 and IEC 62217 relevant to 66 & 132kV Inductive or Capacitive Voltage Transformers for use on the CE Electric UK distribution network.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes. A separate sheet shall be provided for each product being offered.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the Manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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IEC 60044-2 / IEC 60044-5 – Type test requirements			
Clause/Sub-clause	Requirement	Conformance Code	Remarks and or type test report numbers
IEC 60044-5 clause 10.6 only	Accuracy Check		
8.1 / 9.1	Temperature rise test		
8.2 / 9.3	Short-time withstand current tests		
8.3.2 / 9.4.2	Lightning impulse test		
8.3.3	Switching Impulse test		
8.4	Wet test for outdoor type transformers		
12.3 & 13.6.2	Determination of errors - type test for accuracy of measuring current transformers		
IEC 60044-5 clause 9.9 only	Transient response Test		
IEC 60044-5 clause 9.6 only	Ferro-resonance test		
IEC 60044-5 clause 9.7 only	Tightness of electromagnetic unit		

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IEC 60044-2 / IEC 60044-5 – Routine test requirements			
Clause/Sub-clause	Requirement	Conformance Code	Remarks and or type test report numbers
9.1 / 10.3	Verification of terminal markings		
9.2 / 10.2	Power- frequency withstand test on primary windings		
9.2.4 / 10.2.3	Partial Discharge measurement		
9.3 / 10.4.2	Power frequency withstand test on Sec winding		
IEC 60044-5 clause 10.2.4 only	Power frequency withstand test on low voltage terminal		
IEC 60044-5 clause 10.4 only	Power frequency withstand test on the electromagnetic unit		
IEC 60044-2 clause 9.3 only	Power frequency withstand test between sections		
IEC 60044-2 clause 12.4 only	Determination of errors		
IEC 60044-5 clause 10.1 only	Tightness of capacitor voltage divider		
IEC 60044-5 clause 9.2 only	Capacitance and tan measurement at power frequency		

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IEC 61462 – Type tests			IEC 62217			Remarks
Clause / Sub-clause	Requirement	Conformance Code	Clause / Sub-clause	Requirement	Conformance Code	
7.2	<i>Tests on interfaces and connections of end fittings</i>		9.2	<i>Tests on interfaces and connections of end fittings</i>		
7.2.2	Reference dry power frequency test		9.2.3	Reference dry power frequency test		
7.2.3	Thermal Mechanical pre-stressing test		9.2.4	Product specific pre-stressing		
7.2.4	Water Immersion pre-stressing test		9.2.5	Water immersion pre-stressing		
7.2.5	Verification test		9.2.6	Verification test		
7.2.5.1	Visual Examination		9.2.6.1	Visual Examination		
7.2.5.2	Steep Front Impulse Test		9.2.6.2	Steep Front Impulse Test		
7.2.5.3	Dry Power Frequency Voltage Test		9.2.6.3	Dry Power Frequency Voltage Test		
7.3	<i>Tests on shed and housing material</i>		9.3	<i>Tests on shed and housing material</i>		
7.3.1	Hardness Test		9.3.1	Hardness Test		
7.3.2	Accelerated weathering test		9.3.2	Accelerated weathering test		
7.3.3	Tracking and erosion test		9.3.3	Tracking and erosion test		
			9.3.3.1	1000 hour salt fog test		
7.3.4	Flammability Test		9.3.4	Flammability Test		
7.4	<i>Tests on the tube material</i>		9.4	<i>Tests on the core material</i>		
7.4.1	Dye penetration test		9.4.1	Dye penetration test		
7.4.2	Water diffusion test		9.4.2	Water diffusion test		
			Annex A	Wheel test		
			Annex B	5000H - Test at Multiple stresses		

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IEC 61462 – Sample tests			IEC 62217			
Clause / Sub-clause	Requirement	Conformance Code	Clause / Sub-clause	Requirement	Conformance Code	Remarks
9.3.1	Verification of dimensions					
9.4	Mechanical tests					
9.5	Galvanising Test					
9.6	Check of the interface between end fittings and the housing					

IEC 61462 – Routine tests			IEC 62217			
Clause / Sub-clause	Requirement	Conformance Code	Clause / Sub-clause	Requirement	Conformance Code	Remarks
10.2	Visual examination					
10.4	Routine mechanical test					
10.5	Routine tightness test					
11.0	Documentaton					

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Appendix 5: Addendum to supplier requirements

Project-specific installation and protection requirements will be provided by CE Electric UK Primary Engineering Projects for inclusion in this appendix.